

Empirical Proof Record: VALIDATED

2026-05-31T19:26:02.096914+00:00

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Proof ID: b7c76de7201677eba6ab61ff942859b1...

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What this means

This proof tests whether Kimera remembers the ORDER things happened, not just what happened — a filing cabinet (order-blind) versus a landscape you walk, where the path leaves a groove. It does: the same query, after the same documents in a different order, lands differently. Kimera carries the order of experience where retrieval carries only the inventory. (observed order_hysteresis = 0.9444444444444444)

1. Claim

After the same documents in a different ORDER, the same identically-recognised query gets a different prime-address from Kimera (hysteresis) beyond its own run-noise, whereas a set-based retriever returns the identical ranking (order-blind by construction): $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor}) > 0$.

- **Operationalisation:** Run 6 documents + 6 held-out probes through Kimera's entity target (batch) in order A, order A again (determinism control), and order B (shuffled). Per probe: $\text{order_divergence} = 1 - \text{Jaccard}(\text{prime_chain}|A, \text{prime_chain}|B)$; $\text{noise_floor} = 1 - \text{Jaccard}(\text{prime_chain}|A1, \text{prime_chain}|A2)$. $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor})$. Baseline: TF-IDF retrieval order-divergence over the same docs/queries (= 0). VALIDATED iff $\text{order_hysteresis} > 0$ with a significant paired $\text{order_divergence} > \text{noise_floor}$ test.
- **Threshold:** $\text{order_hysteresis} > 0.0 \text{ jaccard_distance}$
- **H0:** $\text{H0: order_hysteresis} \leq 0$ — Kimera's response does not depend on document order beyond run-noise; accumulation is commutative, like a set-based retriever

- **H1:** H1: `order_hysteresis` > 0 — the same query after the same documents in a different order gets a different prime-address; Kimera carries the ORDER of experience (hysteresis) where RAG carries only the inventory

2. Verdict

VALIDATED — observed 0.9444444444444444
`order_hysteresis` +0.9444 = Kimera order-divergence 1.0000 – noise-floor 0.0556 over 6 probes (prime-address); RAG (TF-IDF) order-divergence 0.0000 (set-based, order-blind by construction); paired order>noise Wilcoxon p=0.016 (significant — order matters); manifold-state order-effect coupling $\Delta=0.231499$, mass $\Delta=0.389913$ (noise coupling $\Delta=0.291579$)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.comparison.order_hysteresis	order_hysteresis	0.9444444444444444	—	ophamin 0.11

4. Pre-registration

- Registered at: 2026-05-31T19:26:01.711896+00:00
- Config hash: e9e0949b499b8b4758262350a7ec8186...
- Data hash: 3896f14d8eaa26af9e1b43a0bcd7ece7...

5. Signature

2644a5707cd2d401576ff7fb56d9fb1f3942d3c2c6dbfc215fe7388dd396cf92